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Red blood cell clustering in Poiseuille microcapillary flow

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Red blood cells (RBC) flowing in microcapillaries tend to associate into clusters, i.e., small trains of cells separated from each other by a distance comparable to cell size. This process is usually attributed to slower RBCs acting to create a sequence of trailing cells. Here, based on the first systematic investigation of collective RBC flow behavior in microcapillaries *in vitro* by high-speed video microscopy and numerical simulations, we show that RBC size polydispersity within the physiological range does not affect cluster stability. Lower applied pressure drops and longer residence times favor larger RBC clusters. A limiting cluster length, depending on the number of cells in a cluster, is found by increasing the applied pressure drop. The insight on the mechanism of RBC clustering provided by this work can be applied to further our understanding of RBC aggregability, which is a key parameter implicated in clotting and thrombus formation. © 2012 American Institute of Physics. [<http://dx.doi.org/10.1063/1.4721811>]

INTRODUCTION

Red blood cells (RBCs) perform essential functions, such as oxygen delivery and CO₂ removal from tissues, thanks to their ability to deform and flow in the microvascular network.¹ Thus, understanding how RBC shape and their collective behavior is affected by confined flow geometries is a key physiopathological issue.² Here, we focus on RBC clustering in microcapillary flow, i.e., the formation of trains of RBCs where the distance between consecutive cells is comparable to cell length. RBC clustering has been observed both experimentally, *in vivo* and *in vitro*,^{3–5} and by numerical simulations,^{6,7} and can be regarded as relevant for the well-known process of RBC aggregation, where cells stick to each other piling up in dense 3D structures called *rouleaux* due to some aggregating agent,⁸ such as fibrinogen, whose specific mechanism has not been fully elucidated. In turn, the formation of packed RBC structures plays an important physiological role, being one of the main factors leading to the non-Newtonian behavior of blood, especially to the increase of viscosity at low shear rates. The latter is implicated in clotting and thrombus formation,^{6,9,10} likely due to vascular obstruction. It should be noticed that RBCs can promote platelet aggregation also by chemical signaling, such as the release of ATP¹¹ and ADP under low pO₂, low pH and in response to mechanical deformation.^{10,12} RBC cluster formation has been attributed to the slower cells traveling down the channel, which act to slow down the faster cells, thus creating a train of trailing cells. This effect has been clearly observed in presence of white blood cells,^{5,13} which, being larger and less deformable than RBCs, move at significantly lower speeds in confined geometries and are found at

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the leading front of RBC clusters. However, a conclusive evidence supporting this explanation for RBCs only has not been provided so far. Furthermore in general, several basic questions concerning the physics of RBC clustering under confined flow remain open. What are the most probable cluster sizes for a given flow condition? How does the length of a cluster, having a given number of cells, evolve by varying flow conditions (such as the flow strength)? What is the mechanism of cluster formation? In this work, we address these questions by presenting the first systematic experimental and numerical investigation on human RBC cluster dynamics in microcapillaries.

MATERIALS AND METHODS

The experimental apparatus used in this work has been described in detail elsewhere,¹⁴ and it will be just briefly reviewed here. Fresh venous blood samples, drawn from healthy consenting donors and used within 4 h from collection, are centrifuged to separate RBCs from white cells and platelets. Then the RBCs are resuspended in plasma and diluted to a concentration of 10% by volume with ACD anticoagulant (0.6% citric acid, 1.1% anhydrous dextrose, 2.3% sodium citrate, 96% water) and human albumin. This concentration has been selected in order to have the typical hematocrit H_t value (i.e., the percentage of cell volume with respect to total blood volume) found in microcirculation. The higher values of H_t (around 45%) measured in healthy blood tests are associated with large vessels, whereas in going from the macro- to the microvasculature H_t decreases due to entrance effects and the inhomogeneous radial distribution of RBCs, which tend to concentrate along the centerline (the so called Fahraeus-Lindqvist effect²). Images of the flowing RBCs are acquired by a high speed video camera (Phantom 4.3, operated up to 1000 frames/s) and by using a high magnification oil immersion objective (100x). The experiments are carried out in 10 μm diameter silica microcapillaries (Polymicro Technologies) loaded in a Plexiglass flow cell. The latter is placed on the motorized x-y stage (Ludl) of an inverted microscope (Zeiss Axiovert 100) equipped with a motor assembly for remote focus control (Ludl). The distance between the RBC suspension menisci in two glass reservoirs connected to the flow cell is taken as equal to the pressure drop in the microcapillaries, the other pressure losses being negligible.

RESULTS AND DISCUSSION

We consider a RBC cluster as a cell train of length L in which the gap d between any pair of consecutive cells is equal or lower than $1.5 D$, where D is the characteristic length of a single cell, as shown in Figure 1(a). D is dependent on the imposed pressure drop and ranges between 6 and 10 μm . It can be noticed that RBC shapes inside a cluster are close to the typical parachute-like shapes (either axisymmetric or slipper-like) observed both *in vivo* and *in vitro* on isolated cells^{3,14,15} (Figure 1(a)). RBC clustering is also associated with local fluctuations of H_t , as exemplified in Figure 1(b). This finding is in agreement with qualitative observations of blood flow in microcapillaries *in vivo* and *in vitro*.^{4,7,13}

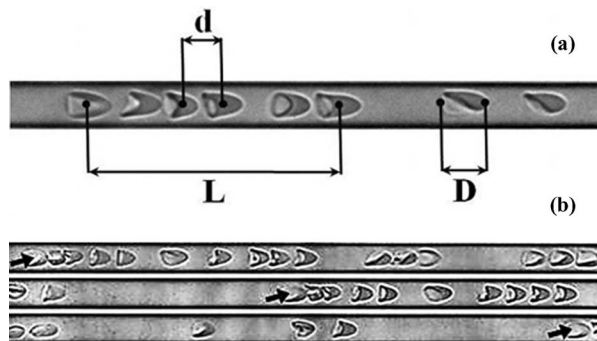


FIG. 1. (a) RBC cluster parameters; (b) microcapillary hematocrit fluctuations (the arrows show the same cell while traveling across the field of view).

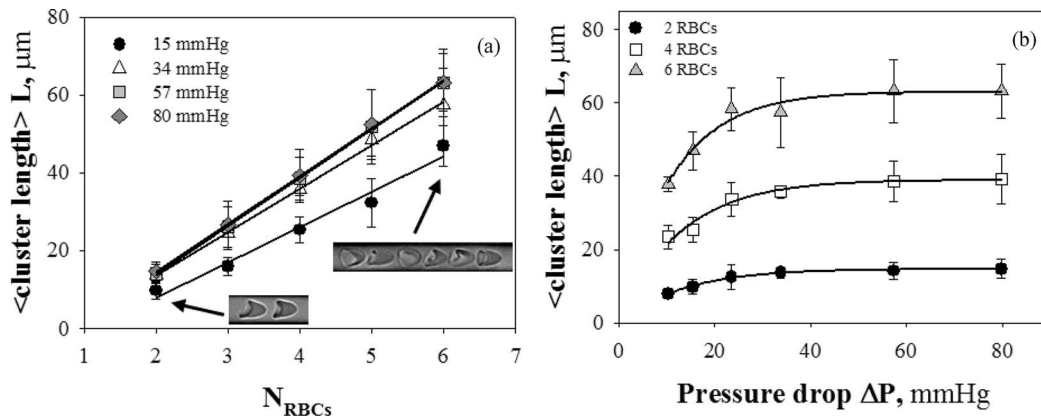


FIG. 2. (a) Clusters length vs. number of RBC per cluster; (b) clusters length vs. ΔP for different numbers of RBCs.

Hence, our experimental model reproduces some important features of RBC flow behavior *in vivo*, both at the single and the collective cell level. The two key experimental parameters in our work are the hematocrit, here fixed at 10%, and the imposed pressure drop ΔP across the capillary of length of about 2 mm. In Figure 2(a), cluster length L is plotted as a function of the number N_{RBC} of RBCs per cluster at four values of ΔP close to the physiological range in microcirculation, which is 15–35 mmHg¹⁶ at the capillary level. A linear increase of L with N_{RBC} , which is well fit by the continuous lines in Figure 2(a), is found at any ΔP . The slope of each linear fit increases with ΔP until a plateau at higher pressure drops is reached, where data superimpose on each other at each value of N_{RBCs} . Error bars (such as in the following figures) represent the standard deviation of the measurements, which is mainly due to the size distribution of RBCs. Thus, there is a critical value of ΔP (which can be estimated to be of about 50 mmHg) above which L remains the same for each N_{RBCs} .

This trend is also shown in Figure 2(b), where L is plotted as a function of ΔP . It can be seen that L increases with N_{RBC} and ΔP up to a plateau level, suggesting the presence of a critical length. We can hypothesize that such plateau level corresponds to the fact that the average intercellular distance d in a cluster stays constant beyond a critical ΔP value even though deformation of isolated RBCs increases.^{14,17} This limiting value of d can be estimated as equal to $L/(N_{\text{RBC}} - 1)$, and is equal to 14.7, 13.1, and 12.6 μm for $N_{\text{RBC}} = 2, 4$, and 6, respectively. To prove this hypothesis, we measured the actual average distance d between the centers of mass of consecutive cells in a cluster and RBC deformation in a cluster. The results are presented in the following of this section. In Table I, the relative distances d_{I-J} between consecutive cells in a cluster (e.g., d_{I-II} corresponds to the distance between the first and the second cell) are tabulated as a function of the number of cells N_{RBCs} forming the cluster and three different pressure drops. For the sake of brevity, only the relative distances of the first four cells in a cluster are reported.

TABLE I. Average distance d_{I-J} (μm) between consecutive cells in a cluster as a function of the number of cell N_{RBCs} forming the cluster at three different ΔP .

N_{RBCs}	10 mmHg			34 mmHg			80 mmHg		
	d_{I-II}	d_{II-III}	d_{III-IV}	d_{I-II}	d_{II-III}	d_{III-IV}	d_{I-II}	d_{II-III}	d_{III-IV}
2	8.3 ± 1.2			11.1 ± 1.5			14.2 ± 2.2		
3	8.4 ± 1.3	8.4 ± 1.8		11.4 ± 1.4	10.8 ± 1.9		13.7 ± 2.8	14 ± 3	
4	8.4 ± 1.4	8.6 ± 2.1	9.7 ± 2.3	11.7 ± 2.4	10.9 ± 2.8	12.1 ± 2.6	14.2 ± 2.6	13.1 ± 3.5	13.3 ± 3.3
5	8.6 ± 2	8.9 ± 3	9.1 ± 2.4	12 ± 2.6	11.2 ± 2.4	11.4 ± 2.4	13.6 ± 2.9	13.5 ± 2.7	12.3 ± 3.3
6	8.8 ± 1.9	8.4 ± 2.6	8.4 ± 2.4	11.5 ± 2.2	10.9 ± 3	11.3 ± 2.6	12.9 ± 2.8	12.8 ± 2.8	12.5 ± 2.7
7	8.7 ± 1.4	8.6 ± 2.9	8.5 ± 1.3	11.6 ± 3.5	12.1 ± 2.5	11.6 ± 3.1	13.4 ± 2.6	11.8 ± 3.3	13.8 ± 2.7

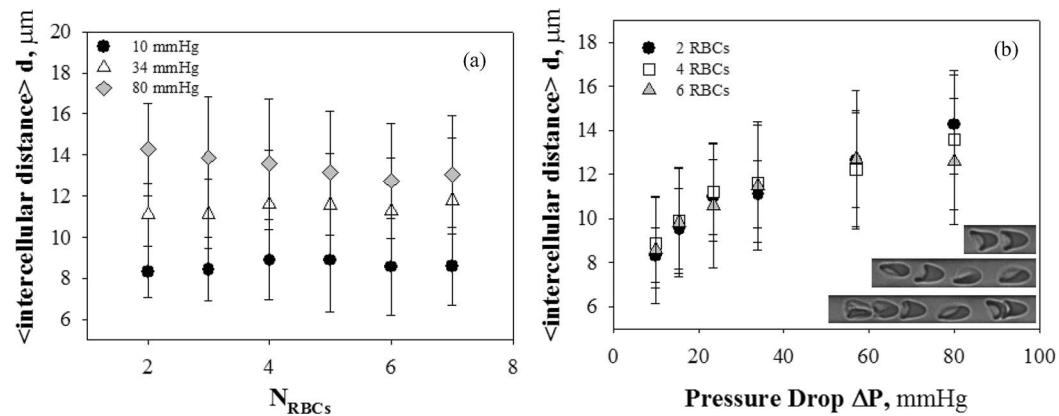


FIG. 3. (a) Distance between consecutive cells in clusters vs. number of RBC per cluster; (b) RBC distance in a cluster vs. ΔP (the images in the inset show clusters of 2, 4, and 6 RBCs at $\Delta P = 10$ mmHg).

The data in Table I show that the distance $d_{i,j}$ increases with ΔP at a given N_{RBCs} , while there is no significant variation with cell position in the cluster and with the number of cells N_{RBCs} composing the cluster. In Figure 3(a), the average intercellular distance d (calculated as the average value of the distances $d_{i,j}$ for each N_{RBCs} ; e.g., for $N_{\text{RBCs}} = 4$ d is the average value of $d_{\text{I-II}}$, $d_{\text{II-III}}$, $d_{\text{III-IV}}$) is plotted as a function of N_{RBCs} at three different values of the imposed pressure drop. In this graph, the independence of d on N_{RBCs} is apparent for the two lower values of ΔP (10 mmHg and 34 mmHg).

For the higher pressure drop (80 mmHg), instead, a weak decrement of d for big clusters ($N_{\text{RBCs}} \geq 5$) is noticed (though within the error bars), suggesting a possible compaction of big clusters at high ΔP . The same trend is shown by the points at the higher ΔP in Figure 3(b), where the average distance d between two consecutive cells in a cluster is plotted as a function of pressure drop for clusters of different N_{RBCs} , showing that d increases with ΔP , apparently reaching a plateau around 60 mmHg, equal to 14.2, 13.6, and 12.7 μm for $N_{\text{RBC}} = 2, 4$, and 6, respectively. These values are quite close to the ones estimated from Figure 2(b), thus supporting our hypothesis. Based on the possible role that RBC deformation could play in the formation, size, and stability of a cluster, a qualitative analysis of RBC shapes is performed by observing the images in the inset of Figure 3(b). Here, three typical examples of clusters at the lower ΔP (10 mmHg) are shown to further illustrate that d is independent of cell position in the cluster and of N_{RBCs} . Another interesting feature of these pictures is that cells of different shapes, such as parachute and slipper-like, are present in a cluster, but there is not a preferential position for each kind of shape (e.g., slipper-like morphologies can be found at the first, at the second, and at the third position as well). To better understand the role of shape in RBC cluster, a quantitative analysis of RBC shape in five cell clusters is presented in Figure 4, where the deformation index DI (measured as the aspect ratio of a bounding box enclosing the cell body) is plotted as a function of pressure drop at different cell positions (e.g., I RBC means the first cell in the cluster starting from the right). The flow is from left to right. Error bars represent the standard deviation of the measurements. An increasing trend is observed, as found for isolated cells,¹⁴ and no remarkable variations of cell shape with position in the clusters are found. The pictures in the inset of Figure 4, which show two examples of five RBC clusters, illustrate how clusters of equal length can exhibit cells of different deformation index at different positions within the cluster. In conclusion, our analysis of cell distance and shape within a cluster provides support to the hypothesis, that the main variable affecting intercellular separation and RBC deformation is ΔP , which drives cluster dimension to a plateau, whose size is dependent on N_{RBCs} (as shown in Figure 2(b)). The presence of a limiting cluster length suggests that the forces keeping the cells together in a cluster are of hydrodynamic nature. This point will be further discussed in the following.

To further investigate cluster formation, size, and stability, the number of RBCs in a cluster as a function of ΔP was measured at three different positions along the capillary: inlet, center, and outlet,

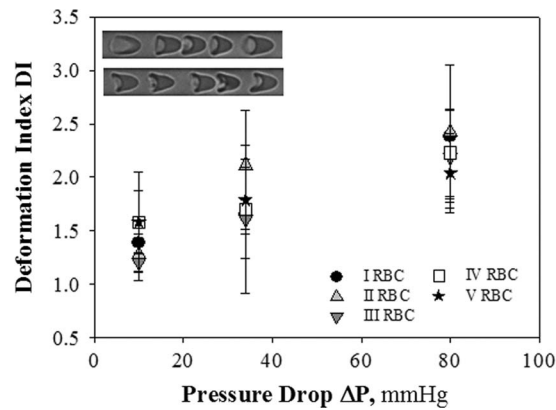


FIG. 4. Plot of RBC deformation index as a function of ΔP in five cell clusters. Typical images are shown in the inset.

as shown in Figure 5. Here, the inlet and the outlet sections are at a distance of $100\ \mu\text{m}$ from the capillary ends (the whole length of the capillary was about $2\ \text{mm}$). At each position the distributions of N_{RBC} are shown at three values of ΔP (15, 34, and $80\ \text{mmHg}$).

At the capillary inlet the average RBC number in a cluster is almost the same independently on ΔP . This is also shown in Table II, where the average of the distribution, which is given as a

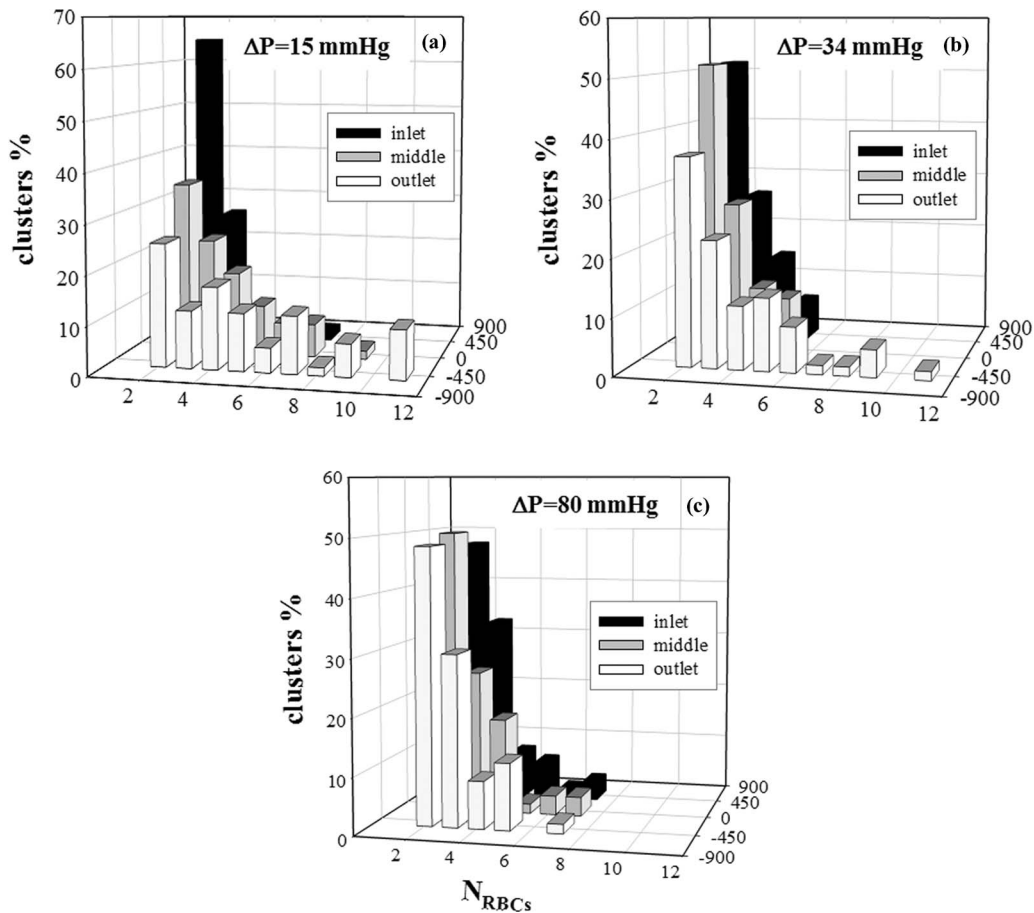


FIG. 5. Distribution of N_{RBC} (the number of RBCs in a cluster) at different positions along the capillary and at the following values of ΔP : (a) $\Delta P = 15\ \text{mmHg}$; (b) $\Delta P = 34\ \text{mmHg}$; (c) $\Delta P = 80\ \text{mmHg}$.

TABLE II. Average number of RBCs in a cluster as a function of pressure drop.

	10 mmHg	34 mmHg	80 mmHg
Inlet	2.5	2.7	2.9
Middle	3.6	2.8	2.9
Outlet	5.2	3.8	2.9

function of ΔP and position, is between 2.5 and 2.9 at the capillary inlet. A possible explanation of this finding is that due to entrance effects RBC clusters are just building up at the inlet section.

On the other hand, the size distributions in the middle and outlet sections are significantly affected by the applied pressure drop. The cluster size distributions become more narrow and the average values lower by increasing ΔP (see also Table II). Thus, at the lower value of ΔP , larger RBC clusters are found in the middle and at the exit of the capillary. Furthermore, at $\Delta P = 15$ and 34 mmHg, the average cluster size increases from the inlet to the outlet section, and this phenomenon could be explained by the relative cell stiffness resisting shear deformations or by the resulting difference in cell shapes at different ΔP . This effect becomes negligible at $\Delta P = 80$ mmHg, where cluster size is the same at each position since pressure drop exceeds the critical ΔP value shown in Figure 2. Finally, measurements of RBC (both isolated and within a cluster) velocity in the 10 μm diameter capillaries (the details of the experimental procedure are described elsewhere¹⁴) have been also carried out in this work. The results show that there is no significant dependence of RBC velocity on cluster size. Furthermore, the difference between cluster and isolated RBC velocity is within 1%.

In order to elucidate the physical mechanisms leading to RBC clustering, we have performed numerical simulations in two dimensions in unbounded Poiseuille flow, and modeled RBCs as vesicles, taking into account bending modulus only (in 2D RBCs and vesicles are equivalent due to the lack of shear elasticity). The simulations are based on the boundary integral method.¹⁸ The Stokes and continuity equations are solved together through the following integral equation for the velocity of the points $\mathbf{x}_0$ of the membrane:

$$\mathbf{u}(\mathbf{x}_0) = \mathbf{u}^\infty(\mathbf{x}_0) + \frac{1}{4\pi\mu} \oint_{\Sigma_i \gamma} \mathbf{G}(\mathbf{x} - \mathbf{x}_0) \cdot \mathbf{f}(\mathbf{x}) ds(\mathbf{x}),$$

where μ is water viscosity, $\mathbf{f}(\mathbf{x})$ is the membrane force, accounting for bending and inextensibility as

$$\mathbf{f} = -k \left[\frac{d^2 c}{ds^2} + \frac{1}{2} c^3 \right] \mathbf{n} + \theta c \mathbf{n} + \frac{d\theta}{ds} \mathbf{t}$$

and $\mathbf{G}$ is a Green's function given by

$$\mathbf{G}_{ij}(\mathbf{x} - \mathbf{x}_0) = -\delta_{ij} \ln |\mathbf{x} - \mathbf{x}_0| + \frac{(\mathbf{x} - \mathbf{x}_0)_i (\mathbf{x} - \mathbf{x}_0)_j}{|\mathbf{x} - \mathbf{x}_0|^2}.$$

Here, $\mathbf{n}$ and $\mathbf{t}$ are the normal and tangential vectors, c the membrane curvature, k the bending modulus and θ the Lagrange multiplier associated with the inextensibility constraint. γ_i represents the surface

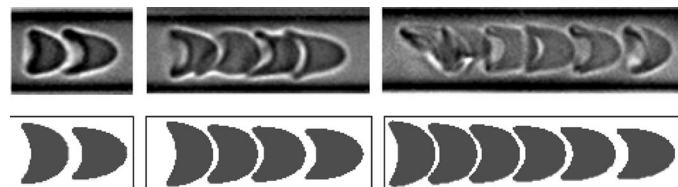


FIG. 6. Comparison of numerical simulations with experimental images of RBCs at the same flow conditions.

TABLE III. Minimal value of R^* as a function of Ca and N .

N	R^*		
	$Ca = 10$	$Ca = 100$	$Ca = 1000$
2	0.94	0.94	0.94
4	0.96	0.95	0.95
6	0.98	0.97	0.96

of the i th vesicle. In order to simulate RBC behavior, the vesicle reduced area (i.e., the ratio between the area of the vesicle and the area of a circle of equal perimeter) is fixed to 0.7, which is the average value for RBCs. Computed clusters of 2, 4, and 6 vesicles are shown in Figure 6 next to the experimental images of RBCs for similar values of the imposed pressure drop.

The corresponding values of Ca , which is the dimensionless capillary number, defined as $\mu v_{\max} r^2/k$ ¹⁹ (where μ is water viscosity, $v_{\max}$ is the Poiseuille maximum velocity, r is the tube radius and k is the RBC bending rigidity, taken as 10^{-19} N/m²⁰), are around 1000, both for experiments and simulations. It can be seen that the predicted cluster morphology bears a quite appealing resemblance with that observed in the experiments. The numerical simulations show that vesicles, at the same reduced area as RBCs, form stable clusters due to the sole hydrodynamical interactions, since our calculations are based on equally sized RBCs. However, it has been often evoked in literature⁴ that clusters form due to size polydispersity of RBCs. Since the largest cells are endowed with the lower velocities,¹⁴ they would slow down the faster moving cells, thus eliciting cell clustering. Our experimental data of single cell and cluster velocity provide a strong evidence against the above-mentioned hypothesis of cluster formation just based on a slowing down effect. Nevertheless, in order to test possible effects of RBC size polydispersity, we have numerically investigated cluster stability upon a variation of the size of vesicles in the less favorable situation, i.e., when a small RBC-like vesicle is in the front position. In fact, due to its reduced size, it would, in principle, move faster than the neighbors, except if it is bound to the cluster by hydrodynamical interactions. We observed that the clusters maintain their stability within a certain variation of the mean radius R of the first vesicle with respect to the radius R_0 of the others. On the other side, the width of this range decreases with increasing vesicle number in the cluster, as indicated in Table III, where N is the number of vesicles and R^* is the minimal R/R_0 allowing the RBCs to form one single cluster.

These results are interesting inasmuch as the standard deviation of the distribution of RBCs radii is around 4% (11%–15% in volume), which is indeed in agreement with values from the literature (the standard deviation of cell volume has been reported as equal to 16%²¹). In fact, the data in Table III allow to predict the existence of unstable RBC clusters for the parameters (N , Ca) corresponding to $R^* > 0.96$. The vesicle shapes for the three configurations corresponding to the limiting cases of Table III at $Ca = 1000$ are represented in Figure 7.

To gain further insight about the mechanisms governing cluster formation, we have analyzed the velocity field around the vesicles. This analysis reveals that a single vesicle in a Poiseuille flow generates a hydrodynamic disturbance formed by two recirculation vortices: one clockwise in the lower half-plane (vesicles move from right to left), the other counterclockwise in the upper half-plane (the detailed results are not shown here for the sake of brevity). The two-dimensional numerical results presented here compare nicely to the experimental observations. Nevertheless, these results should be confirmed by three-dimensional simulations.



FIG. 7. Stable clusters with a smaller vesicle at the front (in dark grey).

CONCLUSIONS

In this work, we show both by experiments and numerical simulations that RBC clustering in microconfined Poiseuille flow is due to hydrodynamic interactions between consecutive cells. In addition, we show that RBC size polydispersity within the physiological range does not affect cluster stability. This work is relevant to improve our understanding of microvascular events where RBC clustering effects can be played a significant role, such as the process of thrombus formation in microcirculation. More in general, our results can be of interest in the framework of confined multiphase flow, where deformable particles, such as droplets, are investigated in microfluidics conditions.²²

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